

0.1 Sequences and Series of Functions: Pointwise vs. Uniform Convergence

The transition from analyzing individual functions to sequences of functions $(f_n)_{n=1}^{\infty}$ requires a careful distinction between local limits and global limits. The mode of convergence dictates which analytic properties—such as continuity, differentiability, and integrability—are preserved in the limit function.

Let $E \subset \mathbb{R}$, and let $(f_n)_{n=1}^{\infty}$ be a sequence of functions defined on E . We say that (f_n) **converges pointwise** to a function $f : E \rightarrow \mathbb{R}$ if, for every $x \in E$ and every $\epsilon > 0$, there exists an integer $N \in \mathbb{N}$ (dependent on both x and ϵ) such that for all $n \geq N$,

$$|f_n(x) - f(x)| < \epsilon$$

Pointwise convergence is often too weak to preserve analytic properties; a limit of continuous functions may be entirely discontinuous. To guarantee the preservation of continuity, we require a stronger, global form of convergence.

A sequence of functions $(f_n)_{n=1}^{\infty}$ **converges uniformly** to f on E if, for every $\epsilon > 0$, there exists an integer $N \in \mathbb{N}$ (dependent only on ϵ) such that for all $n \geq N$ and all $x \in E$,

$$|f_n(x) - f(x)| < \epsilon$$

[Uniform Convergence and Continuity] Let (f_n) be a sequence of continuous functions on $E \subset \mathbb{R}$. If (f_n) converges uniformly to f on E , then f is continuous on E .

Let $c \in E$ and let $\epsilon > 0$ be given. We must find a $\delta > 0$ such that for all $x \in E$ with $|x - c| < \delta$, we have $|f(x) - f(c)| < \epsilon$.

By the uniform convergence of (f_n) , there exists $N \in \mathbb{N}$ such that for all $x \in E$,

$$|f_N(x) - f(x)| < \frac{\epsilon}{3}$$

Because f_N is continuous at c , there exists a $\delta > 0$ such that for all $x \in E$ satisfying $|x - c| < \delta$,

$$|f_N(x) - f_N(c)| < \frac{\epsilon}{3}$$

We apply the triangle inequality to bound the difference $|f(x) - f(c)|$ for any $x \in E$ strictly within δ of c :

$$|f(x) - f(c)| \leq |f(x) - f_N(x)| + |f_N(x) - f_N(c)| + |f_N(c) - f(c)|$$

Substituting our bounds yields:

$$|f(x) - f(c)| < \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon$$

Thus, f is continuous at c . Since c was arbitrary, f is continuous on the entirety of E .

When analyzing infinite series of functions, $\sum_{n=1}^{\infty} f_n(x)$, uniform convergence is most efficiently established via a bounding criterion over the summands.

[Weierstrass M-Test] Let (f_n) be a sequence of functions defined on E , and suppose there exists a sequence of real numbers (M_n) such that $|f_n(x)| \leq M_n$ for all $x \in E$ and all $n \in N$. If the infinite series $\sum_{n=1}^{\infty} M_n$ converges, then the series $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly on E .

Let $S_k(x) = \sum_{n=1}^k f_n(x)$ denote the sequence of partial sums. We will show that (S_k) satisfies the uniform Cauchy criterion. Let $\epsilon > 0$. Because $\sum_{n=1}^{\infty} M_n$ converges, the sequence of its partial sums is Cauchy. Thus, there exists an $N \in N$ such that for all $m > k \geq N$,

$$\sum_{n=k+1}^m M_n < \epsilon$$

For any $x \in E$ and for $m > k \geq N$, the triangle inequality implies:

$$|S_m(x) - S_k(x)| = \left| \sum_{n=k+1}^m f_n(x) \right| \leq \sum_{n=k+1}^m |f_n(x)| \leq \sum_{n=k+1}^m M_n < \epsilon$$

Since N depends strictly on ϵ and is entirely independent of x , the sequence of partial sums (S_k) is uniformly Cauchy. Therefore, the series converges uniformly to a limit function on E .

[Integration of Uniformly Convergent Sequences] Let α be monotonically increasing on $[a, b]$. Suppose $f_n \in \mathcal{R}(\alpha)$ on $[a, b]$ for $n = 1, 2, \dots$, and suppose $f_n \rightarrow f$ uniformly on $[a, b]$. Then $f \in \mathcal{R}(\alpha)$ on $[a, b]$, and:

$$\int_a^b f d\alpha = \lim_{n \rightarrow \infty} \int_a^b f_n d\alpha$$

Let $\epsilon > 0$. Choose a constant $\epsilon_0 > 0$ such that $\epsilon_0(\alpha(b) - \alpha(a)) < \frac{\epsilon}{3}$. By the definition of uniform convergence, there exists $N \in N$ such that for all $n \geq N$ and $x \in [a, b]$,

$$f_n(x) - \epsilon_0 < f(x) < f_n(x) + \epsilon_0$$

This strict bounding implies that for the upper and lower Darboux-Stieltjes integrals:

$$\overline{\int} f d\alpha \leq \overline{\int} (f_n + \epsilon_0) d\alpha = \int_a^b f_n d\alpha + \epsilon_0(\alpha(b) - \alpha(a))$$

$$\underline{\int} f d\alpha \geq \underline{\int} (f_n - \epsilon_0) d\alpha = \int_a^b f_n d\alpha - \epsilon_0(\alpha(b) - \alpha(a))$$

Therefore, the difference between the upper and lower integrals of f is strictly bounded by:

$$0 \leq \overline{\int} f d\alpha - \underline{\int} f d\alpha \leq 2\epsilon_0(\alpha(b) - \alpha(a)) < \frac{2\epsilon}{3} < \epsilon$$

Since ϵ was arbitrary, the upper and lower integrals coincide: $\overline{\int} f d\alpha = \underline{\int} f d\alpha$, proving $f \in \mathcal{R}(\alpha)$. Furthermore, evaluating the difference of the integrals for $n \geq N$ gives:

$$\left| \int_a^b f d\alpha - \int_a^b f_n d\alpha \right| = \left| \int_a^b (f - f_n) d\alpha \right| \leq \int_a^b |f - f_n| d\alpha \leq \epsilon_0(\alpha(b) - \alpha(a)) < \epsilon$$

This proves that the sequence of integrals converges to the integral of the limit function.